

ALTURAS, CA, USA

WMO: 725958

Lat: 41.483N

Lon: 120.567W

Elev: 4375

StdP: 12.52

Time Zone: -8.00 (NAP)

Period: 98-19

WBAN: 94299

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	3.5	10.1	-1.1	6.1	11.4	4.6	8.2	19.1	25.8	41.6	22.0	39.3	0.9	20	0.601

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	41.0	94.4	60.0	91.4	59.1	88.9	58.1	62.5	87.5	60.8	85.7	59.5	84.4	9.3	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
54.1	73.2	66.7	51.4	66.1	66.4	48.4	59.2	64.6	30.1	87.4	28.8	85.7	27.8	84.3	69.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.1	19.0	16.9	-5.6	99.5	8.2	2.6	-11.5	101.4	-16.3	102.9	-20.8	104.3	-26.8	106.2
			-5.9	65.4	8.0	2.2	-11.7	67.0	-16.4	68.2	-20.9	69.4	-26.7	71.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	47.7	31.8	34.2	39.2	44.0	52.3	60.6	68.4	66.0	58.4	47.4	37.8	30.8	(6)
	DBStd	14.56	8.49	7.01	6.45	6.80	6.97	6.73	5.05	4.74	6.60	6.59	7.87	8.19	(7)
	HDD50	2709	565	442	336	201	57	7	0	0	12	126	368	595	(8)
	HDD65	6575	1030	862	799	631	396	158	23	43	211	547	815	1060	(9)
	CDD50	1854	0	0	2	20	127	326	571	497	263	45	3	0	(10)
	CDD65	245	0	0	0	0	1	28	129	75	12	0	0	0	(11)
	CDH74	9060	0	0	1	24	288	1288	3494	2689	1179	97	0	0	(12)
	CDH80	4060	0	0	0	1	66	484	1797	1254	438	20	0	0	(13)
(14) Wind	WSAvg	5.3	4.3	5.3	6.4	6.7	6.5	6.3	5.7	5.1	4.5	4.4	4.4	4.6	(14)
(15) Precipitation	PrecAvg	14.1	1.6	1.3	1.6	1.6	1.7	0.9	0.2	0.3	0.4	1.0	1.6	1.9	(15)
	PrecMax	23.5	3.9	3.1	4.2	3.9	5.6	2.4	0.8	1.2	1.4	3.1	4.2	5.3	(16)
	PrecMin	8.3	0.1	0.4	0.6	0.4	0.0	0.1	0.0	0.0	0.0	0.0	0.2	0.5	(17)
	PrecStd	4.4	1.1	0.7	0.9	0.8	1.4	0.7	0.2	0.4	0.4	0.8	0.9	1.5	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.7	61.3	70.4	77.3	86.9	94.5	99.3	96.7	92.5	83.3	69.5	54.0	(19)
		MCWB	43.1	45.5	48.7	52.3	57.9	60.6	63.0	61.0	58.8	54.5	49.6	42.8	(20)
	2%	DB	53.1	55.4	64.2	72.3	81.4	89.6	95.4	93.0	88.3	75.5	63.7	49.9	(21)
		MCWB	41.9	42.1	46.3	50.3	55.9	59.0	60.3	59.6	56.4	51.6	47.5	41.0	(22)
	5%	DB	48.4	51.1	59.9	67.3	77.0	85.6	92.7	90.3	85.2	72.4	57.4	46.5	(23)
		MCWB	40.2	41.3	44.3	48.1	53.5	56.8	59.5	58.5	55.5	50.0	44.4	40.2	(24)
	10%	DB	44.8	46.7	54.4	62.3	72.3	81.8	90.1	87.6	81.7	67.2	52.4	43.5	(25)
		MCWB	38.7	39.0	42.4	45.9	51.7	55.9	58.9	57.4	54.8	48.2	42.6	38.5	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.1	47.4	50.2	54.3	59.9	62.9	65.4	64.9	62.0	55.6	51.3	45.6	(27)
		MCDB	54.6	57.9	66.3	72.6	81.8	88.4	92.5	91.4	81.7	75.4	65.6	49.9	(28)
	2%	WB	43.5	44.3	47.6	51.4	57.0	60.1	63.2	61.6	59.2	53.1	48.4	43.2	(29)
		MCDB	50.0	52.9	61.1	69.5	77.9	85.0	89.1	86.2	81.6	72.0	60.7	47.7	(30)
	5%	WB	41.0	41.8	45.5	49.1	54.9	58.4	61.4	59.9	57.0	51.2	46.0	41.2	(31)
		MCDB	47.1	49.1	57.7	64.9	73.4	82.0	86.9	84.7	81.0	69.3	56.2	45.3	(32)
	10%	WB	38.9	39.6	43.3	46.8	52.7	56.7	59.9	58.5	55.2	49.2	43.6	38.8	(33)
		MCDB	44.2	46.5	53.4	60.2	69.3	78.8	85.7	83.4	79.7	64.9	51.3	43.1	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	22.3	22.2	25.1	27.1	30.1	35.5	41.0	41.3	40.3	34.6	25.8	20.7	(35)
		MCDBR	29.6	30.1	37.8	40.8	41.0	43.4	44.5	45.7	47.6	45.6	36.8	25.9	(36)
		MCWBR	19.4	18.6	21.0	20.7	18.2	16.8	16.5	18.7	21.6	23.9	22.1	17.9	(37)
	5% WB	MCDBR	24.2	25.5	33.1	37.1	36.9	40.4	41.0	41.3	43.6	40.2	32.4	19.3	(38)
		MCWBR	16.4	16.3	19.0	19.2	16.7	16.3	15.8	17.7	20.4	21.8	19.7	13.8	(39)
(40) Clear-Sky Solar Irradiance	taub		0.279	0.289	0.300	0.307	0.314	0.308	0.312	0.310	0.298	0.297	0.280	0.275	(40)
	taud		2.507	2.510	2.505	2.493	2.513	2.557	2.545	2.530	2.560	2.551	2.563	2.531	(41)
	Ebn at Noon		281	295	303	306	305	305	303	301	300	288	280	274	(42)
	Edh at Noon		23	27	30	33	33	32	32	31	28	25	22	21	(43)
(44) All-Sky Solar Radiation	RadAvg		627	912	1288	1723	2133	2501	2536	2243	1784	1194	709	502	(44)
	RadStd		83	113	123	167	190	131	108	106	89	100	75	98	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (7 neighbors)	N/A	N/A	N/A	+1.06	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	+1.08	N/A	N/A	N/A	N/A	+164	+76

Nomenclature: See separate page